

F100 / WATER FILTER

Seven components. One pressure-boundary weld.

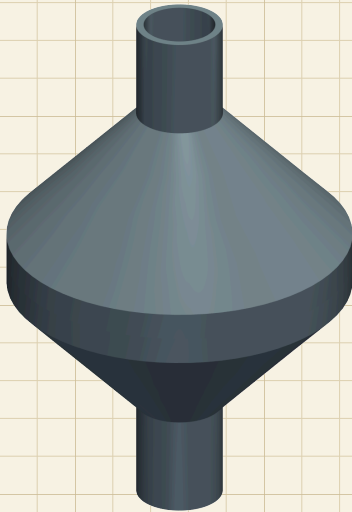
DESIGN INTENT

[insert text here]

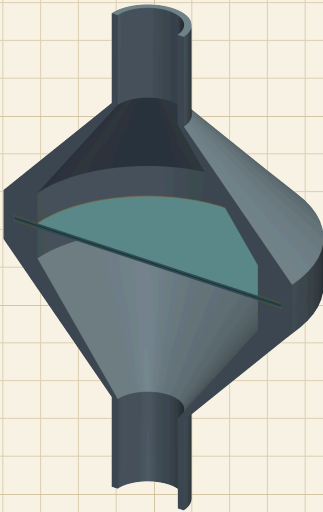
PROJECT	Water filtration
DOCUMENT	F100-SIMPLE-001
AUTHOR	Engineering concept
CHECKED	Review required
REV	S0
DATE	2026-09-14

F100 / ASSEMBLED ISOMETRIC AND CUTAWAY

ASSEMBLED ISOMETRIC



HOUSING AND PACK / CUTAWAY ISO



CALCULATION CONSTANTS

Edit constants.csv, the editable source of this table, then run regenerate.py. All dimensional and engineering inputs below are read from that table. Derived CAD volume is regenerated from those dimensions. Changing temperature also requires updating the corresponding water properties.

CONSTANTS

Key	Value / unit	Definition / basis
temperature_F	70 degF	Service water temperature; User
rho_w	997.99 kg/m3	Water density at stated temperature; IAPWS approximation; pressure effects omitted
mu_w	0.000979 Pa s	Water dynamic viscosity; IAPWS approximation
mdot_n	0.14 lb/s	Nominal mass flow; User
mdot_s	0.47 lb/s	Surge mass flow; User
p_meop	650 psi	Housing operating pressure difference; User: internal minus external
p_proof	1300 psi	Housing proof pressure difference; User
p_burst	2600 psi	Housing burst pressure difference; User
dp_clean_limit	5 psi	Clean pressure-drop limit; User
dp_loaded_limit	18 psi	Loaded pressure-drop limit; User; nominal flow basis
dirt_mass	0.318 g	Retained dry dirt mass; User; contaminant unspecified
mass_limit	1 lb	Maximum dry hardware mass; User
rating	5 um	Filtration rating target; User; test definition pending
efficiency	99.9 percent	Capture efficiency target; User

CONSTANTS / CONTINUED

Key	Value / unit	Definition / basis
efficiency_size	3 um	Efficiency particle size; User
active_d	58 mm	Frame opening / exposed screen diameter; Design
frame_d	70 mm	Frame and mesh outside diameter; Design
frame_raw_t	0.2 mm	Uncoined sheet thickness; Design
frame_assembled_t	0.17 mm	Nominal coined frame envelope; Unvalidated manufacturing assumption
coarse_t	0.48 mm	Each coarse envelope thickness; Calender target; raw catalog 0.508 mm
fine_t	0.18 mm	Dutch-twill envelope thickness; Placeholder; NOT supplier-confirmed Dutch twill
body_d	76 mm	Housing outside diameter; Simplified design
pocket_d	70.4 mm	Single frame seat outer diameter; Design
outer_shoulder_z	6 mm	Outer cone start from weld plane; Design
inner_shoulder_z	8 mm	Inner cone start from weld plane; Design
stub_start_z	36 mm	Cone end / straight stub start; Design
half_length	58 mm	Housing half length; Design
stub_od	19.05 mm	Integral weld stub outside diameter; Design; compatible titanium lines

CONSTANTS / CONTINUED

Key	Value / unit	Definition / basis
stub_id	15.75 mm	Flow bore diameter; Design
Cv_f	6e+07 1/m	Fine Dutch-twill viscous resistance; Assumed; fit finished cloth data
Ci_f	3000 1	Fine Dutch-twill inertial resistance; Assumed; not transferred supplier data
Cv_c	200000 1/m	Combined coarse viscous resistance; Assumed
Ci_c	40 1	Combined coarse inertial resistance; Assumed
K_body	3 1	Combined housing loss coefficient; Assumed; changed housing not flow-validated
alpha_c	1e+09 m/kg	Specific cake resistance; Sensitivity input; contaminant uncharacterized
rho_ti	4510 kg/m3	CP titanium Grade 2 density; TIMET
rho_ss	8000 kg/m3	Stainless frame density; Engineering nominal
Sy	275 MPa	Ti Grade 2 minimum yield screening value; TIMET / ATI; stock certificate required
Su	345 MPa	Ti Grade 2 minimum tensile screening value; ATI
coarse_areal_mass	1.367 kg/m2	Each coarse cloth areal mass; TWP 0.280 lb/ft2 rounded
fine_areal_mass	0.95 kg/m2	Dutch-twill areal mass; Placeholder; NOT supplier-confirmed Dutch twill
growth_factor	1.2 1	Dry mass development reserve multiplier; Design assumption

01 / REQUIREMENTS AND INTERPRETATION

[insert text here]

REQUIREMENT-BY-REQUIREMENT ASSESSMENT

ID / requirement	Constant / target	Assessment
R01 / Housing MEOP	p_meop / 650 psi	[insert text here]
R02 / Housing proof	p_proof / 1300 psi	[insert text here]
R03 / Housing burst	p_burst / 2600 psi	[insert text here]
R04 / Micron rating	rating / 5 um	[insert text here]
R05 / Efficiency	efficiency / 99.9 percent at 3 um	[insert text here]
R06 / Nominal flow	mdot_n / 0.14 lb/s	[insert text here]
R07 / Surge flow	mdot_s / 0.47 lb/s	[insert text here]
R08 / Clean pressure drop	dp_clean_limit / 5 psi	[insert text here]
R09 / Retained dirt	dirt_mass / 0.318 g	[insert text here]
R10 / Loaded pressure drop	dp_loaded_limit / 18 psi	[insert text here]
R11 / Dry mass	mass_limit / 1 lb	[insert text here]
R12 / Water temperature	temperature_F / 70 degF	[insert text here]
R13 / Component count	Seven	[insert text here]

02 / ARCHITECTURE

[insert text here]

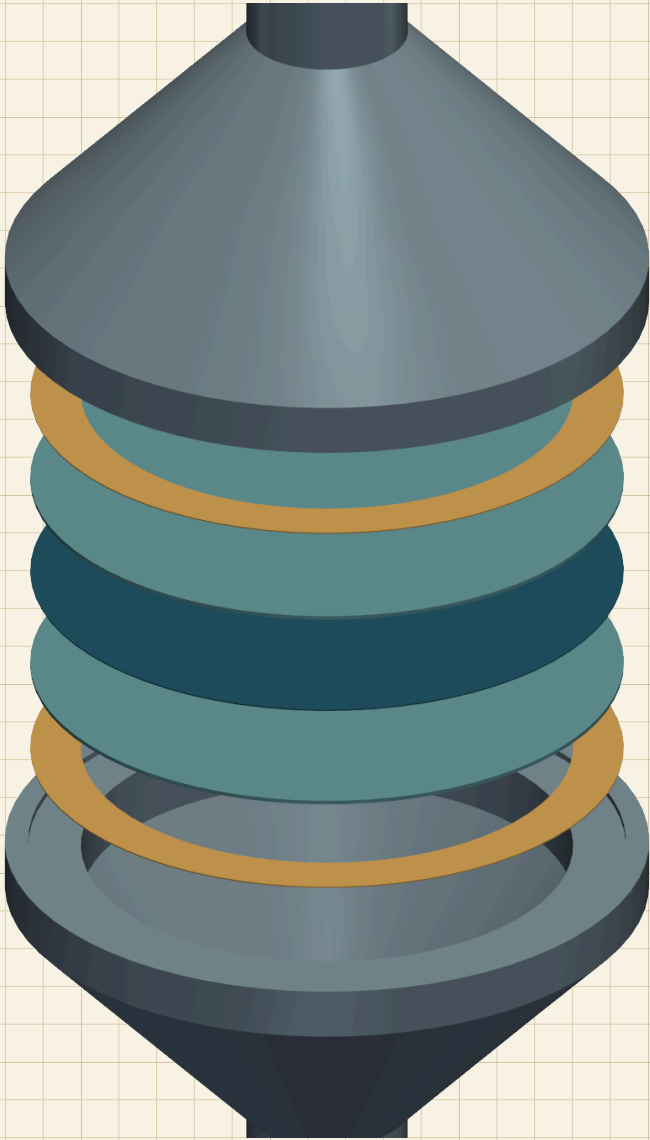
SEVEN-PIECE BILL OF MATERIALS

Component	Qty	Definition
Housing half	2	CP Ti Grade 2; identical; integral weld stubs
Annular frame	2	316L; OD 70, ID 58, raw thickness 0.2 mm
Coarse support	2	40 x 40; assembled thickness 0.48 mm
Fine cloth	1	Dutch twill candidate; thickness 0.18 mm assumed

03 / ASSEMBLY EXPLODED

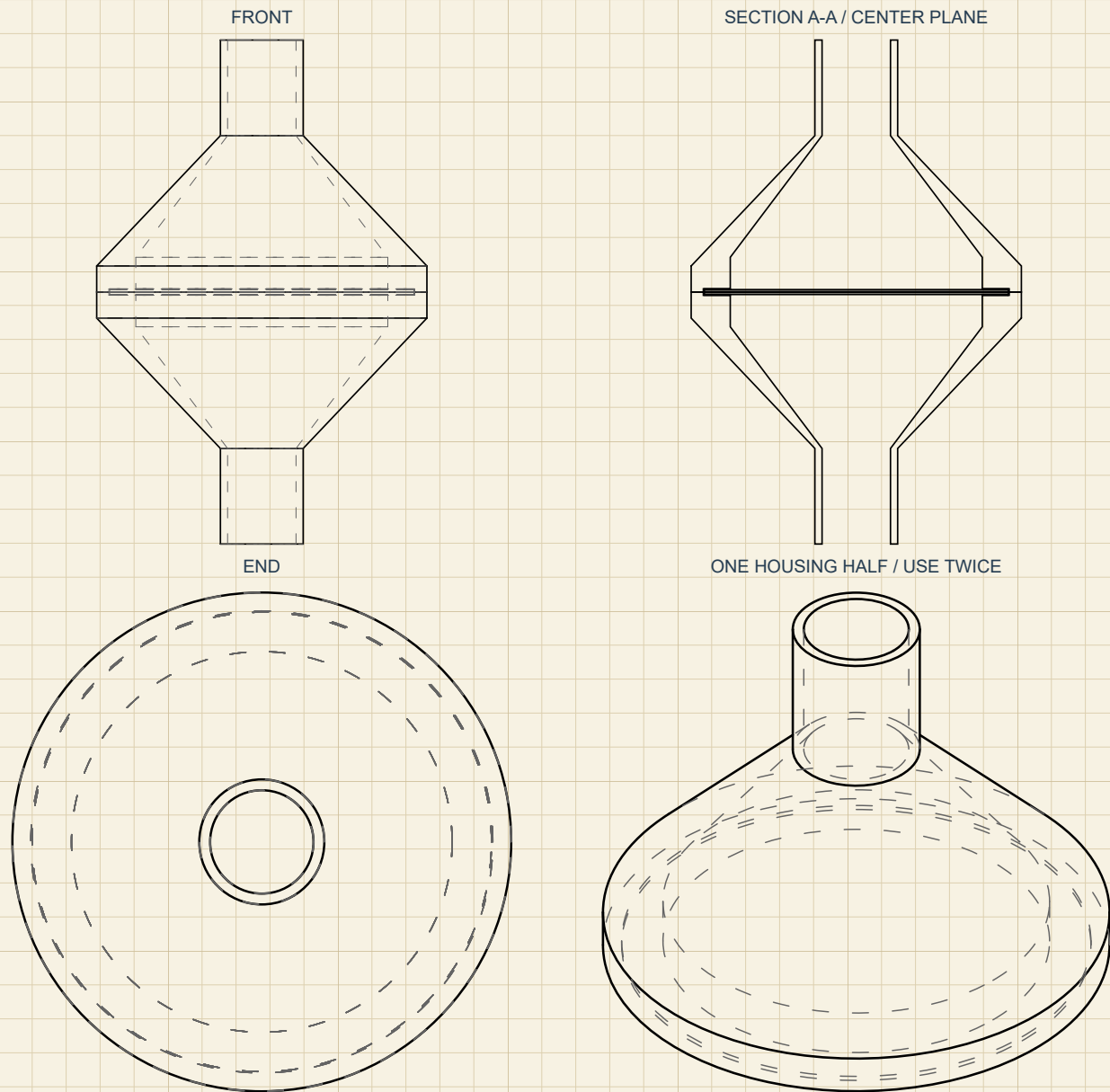
[insert text here]

SEVEN COMPONENTS



[insert text here]

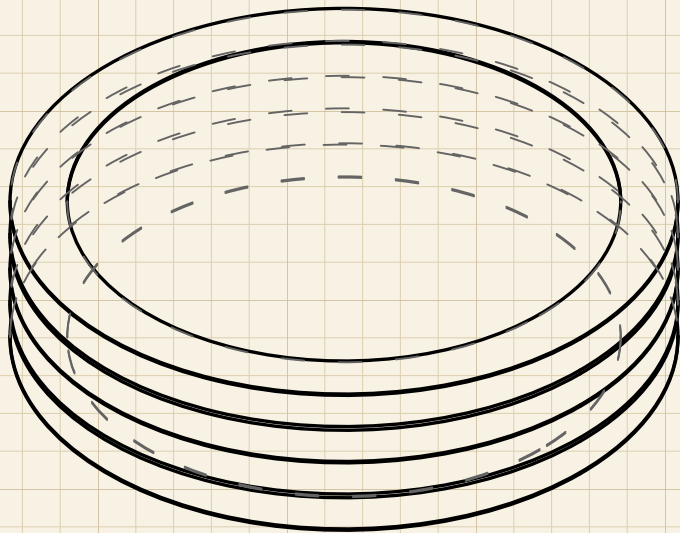
ALL FOUR DRAWINGS



06 / FILTER ELEMENT

[insert text here]

THREE-MESH PACK



Build123d exploded element; media envelopes, not modeled pores.

MESH SOURCING

Supplier / item		Catalog basis	Assessment
GKD / Dutch-twilled family		5 um end of family; 316L listed; exact construction needs RFQ	[insert text here]
TWP / 040X040T0100		Coarse 40 x 40, 0.010 in wire; T316	[insert text here]
Darby / 40316.010PL		Coarse 40 x 40, 0.010 in wire; T316	[insert text here]
Dorstener / custom fabrication		Custom cloth and element fabrication lead	[insert text here]

SOURCE TRAIL

[insert text here]

[2] [3] [4] [5]

08 / FLOW BASIS AND CLEAN-SCREEN EQUATION

[insert text here]

WATER AND GEOMETRY

$\rho_w = 997.990 \text{ kg/m}^3$

$\mu_w = 0.001 \text{ Pa}\cdot\text{s}$

$d_{\text{face}} = 58.000 \text{ mm}$

$d_{\text{bore}} = 15.750 \text{ mm}$

$\dot{m}_n = 0.140 \text{ lb/s}$

$\dot{m}_s = 0.470 \text{ lb/s}$

EXPPOSED AREA AND APPROACH VELOCITY

$$A_{\text{face}} = \pi \cdot \frac{(d_{\text{face}})^2}{4} = 3.142 \cdot \frac{(58.000 \text{ mm})^2}{4} = 2642.079 \text{ mm}^2 \tag{1}$$

$$U_n = \frac{\dot{m}_n}{\rho_w \cdot A_{\text{face}}} = \frac{0.140 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 2642.079 \text{ mm}^2} = 0.024 \frac{\text{m}}{\text{s}} \tag{2}$$

$$U_s = \frac{\dot{m}_s}{\rho_w \cdot A_{\text{face}}} = \frac{0.470 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 2642.079 \text{ mm}^2} = 0.081 \frac{\text{m}}{\text{s}} \tag{3}$$

$$U_s = 0.081 \text{ m/s}$$

SCREEN-SPECIFIC RESISTANCE MODEL

$$\text{dp}_{\text{screen}} = C_i \cdot \rho \cdot U^2 + C_v \cdot U \cdot \mu \tag{4}$$

USE OF THE SUPPLIED EQUATIONS

[insert text here]

$$C_v = A_1 \frac{L}{D_a^2}$$

$$C_i = A_2 \frac{L}{D_a}$$

09 / SIMPLIFIED SURGE ESTIMATION

[insert text here]

HYDRAULIC HYPOTHESES / CONSTANTS TABLE

$$C_{vf} = 60000000.000 \text{ 1/m}$$

$$C_{if} = 3000.000$$

$$C_{vc} = 200000.000 \text{ 1/m}$$

$$C_{ic} = 40.000$$

$$K_{body} = 3.000$$

SURGE ASSEMBLY PRESSURE DROP

$$A_{bore} = \pi \cdot \frac{(d_{bore})^2}{4} = 3.142 \cdot \frac{(15.750 \text{ mm})^2}{4} = 194.828 \text{ mm}^2 \quad (5)$$

$$V_{bore} = \frac{\dot{m}_s}{\rho_w \cdot A_{bore}} = \frac{0.470 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 194.828 \text{ mm}^2} = 1.096 \frac{\text{m}}{\text{s}} \quad (6)$$

$$DP_s = (C_{vf} + C_{vc}) \cdot \mu_w \cdot U_s + (C_{if} + C_{ic}) \cdot \rho_w \cdot (U_s)^2 + K_{body} \cdot \rho_w \cdot \frac{(V_{bore})^2}{2} \quad (7)$$

$$= (60000000.000 \frac{1}{\text{m}} + 200000.000 \frac{1}{\text{m}}) \cdot 0.001 \text{ Pa} \cdot \text{s} \cdot 0.081 \frac{\text{m}}{\text{s}} + (3000.000 + 40.000) \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot (0.081 \frac{\text{m}}{\text{s}})^2 + 3.000 \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot \frac{(1.096 \frac{\text{m}}{\text{s}})^2}{2} \quad (8)$$

$$= 3.829 \text{ psi} \quad (9)$$

$$DP_s = 3.829 \text{ psi}$$

NOMINAL ASSEMBLY PRESSURE DROP

$$V_{nom} = \frac{\dot{m}_n}{\rho_w \cdot A_{bore}} = \frac{0.140 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 194.828 \text{ mm}^2} = 0.327 \frac{\text{m}}{\text{s}} \quad (10)$$

$$DP_n = (C_{vf} + C_{vc}) \cdot \mu_w \cdot U_n + (C_{if} + C_{ic}) \cdot \rho_w \cdot (U_n)^2 + K_{body} \cdot \rho_w \cdot \frac{(V_{nom})^2}{2} \quad (11)$$

$$= (60000000.000 \frac{1}{\text{m}} + 200000.000 \frac{1}{\text{m}}) \cdot 0.001 \text{ Pa} \cdot \text{s} \cdot 0.024 \frac{\text{m}}{\text{s}} + (3000.000 + 40.000) \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot (0.024 \frac{\text{m}}{\text{s}})^2 + 3.000 \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot \frac{(0.327 \frac{\text{m}}{\text{s}})^2}{2} \quad (12)$$

$$= 0.484 \text{ psi} \quad (13)$$

$$DP_n = 0.484 \text{ psi}$$

10 / FIT STRATEGY

[insert text here]

Fine media: Dutch twill. Cv and Ci remain assumed until fitted to the selected finished cloth.

FIT THE SUPPLIED MODEL

[insert text here]

$$\Delta \frac{p}{\mu U} = C_v + C_i \left(\rho \frac{U}{\mu} \right)$$

11 / LOADED PRESSURE DROP

[insert text here]

SERIES-CAKE EQUATION

[insert text here]

$$\Delta p_l = C_v \mu U + C_i \rho U^2 + \mu U \alpha_c \frac{m_d}{A_f}$$

CAKE INPUTS / CONSTANTS TABLE

$\alpha_c = 1000000000.000 \text{ m/kg}$

$m_d = 0.318 \text{ g}$

LOADED ASSEMBLY PRESSURE DROP

$$\text{DP}_{loaded_n} = \text{DP}_n + \mu_w \cdot U_n \cdot \alpha_c \cdot \frac{m_d}{A_{face}} = 0.48424994378907155 \text{pound}_{force_{per_{square}_{inch}}} + 0.001 \text{ Pa} \cdot \text{s} \cdot 0.024 \frac{\text{m}}{\text{s}} \cdot 1000000000.000 \frac{\text{m}}{\text{kg}} \cdot \frac{0.318 \text{ g}}{2642.079 \text{ mm}^2} = 0.896 \text{ psi} \quad (14)$$

$$\text{DP}_{loaded_s} = \text{DP}_s + \mu_w \cdot U_s \cdot \alpha_c \cdot \frac{m_d}{A_{face}} = 3.8286324016148194 \text{pound}_{force_{per_{square}_{inch}}} + 0.001 \text{ Pa} \cdot \text{s} \cdot 0.081 \frac{\text{m}}{\text{s}} \cdot 1000000000.000 \frac{\text{m}}{\text{kg}} \cdot \frac{0.318 \text{ g}}{2642.079 \text{ mm}^2} = 5.210 \text{ psi} \quad (15)$$

$$\text{DP}_{loaded_s} = 5.210 \text{ psi}$$

12 / HOUSING PRESSURE SCREENING

[insert text here]

PRESSURE AND WELD-RIM SECTION

$p_{\text{meop}} = 4.482 \text{ MPa}$
$p_{\text{proof}} = 8.963 \text{ MPa}$
$p_{\text{burst}} = 17.926 \text{ MPa}$
$r_i = 35.200 \text{ mm}$
$r_o = 38.000 \text{ mm}$
$S_y = 275.000 \text{ MPa}$
$S_u = 345.000 \text{ MPa}$

CLOSED-END THICK-CYLINDER CHECK AT THE WELD RIM

$$\sigma_h = p_{\text{burst}} \cdot \frac{(r_o)^2 + (r_i)^2}{(r_o)^2 - (r_i)^2} \tag{16}$$

$$= 17.93 \text{ MPa} \cdot \frac{(38.00 \text{ mm})^2 + (35.20 \text{ mm})^2}{(38.00 \text{ mm})^2 - (35.20 \text{ mm})^2} \tag{17}$$

$$= 234.67 \text{ MPa} \tag{18}$$

$$\sigma_z = p_{\text{burst}} \cdot \frac{(r_i)^2}{(r_o)^2 - (r_i)^2} = 17.93 \text{ MPa} \cdot \frac{(35.20 \text{ mm})^2}{(38.00 \text{ mm})^2 - (35.20 \text{ mm})^2} = 108.37 \text{ MPa} \tag{19}$$

$$\sigma_r = -17.93 \text{ MPa} \tag{20}$$

$$\sigma_{vm} = \left(\frac{(\sigma_h - \sigma_z)^2 + (\sigma_z - \sigma_r)^2 + (\sigma_r - \sigma_h)^2}{2} \right)^{0.5} \tag{21}$$

$$= \left(\frac{(234.67 \text{ MPa} - 108.37 \text{ MPa})^2 + (108.37 \text{ MPa} - -17.93 \text{ MPa})^2 + (-17.93 \text{ MPa} - 234.67 \text{ MPa})^2}{2} \right)^{0.5} \tag{22}$$

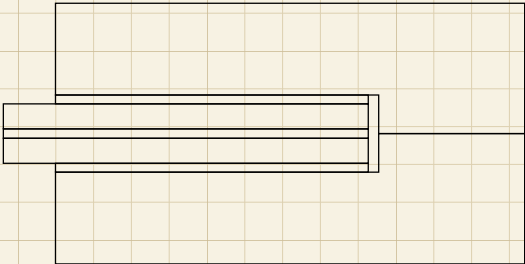
$$= 218.75 \text{ MPa} \tag{23}$$

$\sigma_{vm} = 218.75 \text{ MPa}$

DETAIL B

[insert text here]

HOUSING AND PACK / SECTION DETAIL



Direct Build123d center-plane section; equal geometric scale, no callouts.

22 / FULL MASS CALCULATIONS

[insert text here]

DENSITY, SHEET AND CLOTH INPUTS / CONSTANTS TABLE

$$V_{\text{half}} = 33552.778 \text{ mm}^3$$

$$\rho_{\text{ti}} = 4510.000 \text{ kg/m}^3$$

$$\rho_{\text{ss}} = 8000.000 \text{ kg/m}^3$$

$$d_{\text{pack}} = 70.000 \text{ mm}$$

$$t_{\text{frame}} = 0.200 \text{ mm}$$

$$w_{\text{coarse}} = 1.367 \text{ kg/m}^2$$

$$w_{\text{fine}} = 0.950 \text{ kg/m}^2$$

$$\text{growth} = 1.200$$

$$m_{\text{limit}} = 1.000 \text{ lb}$$

HOUSING, FRAMES AND MESH MASSES

$$A_{\text{pack}} = \pi \cdot \frac{(d_{\text{pack}})^2}{4} = 3.142 \cdot \frac{(70.000 \text{ mm})^2}{4} = 3848.451 \text{ mm}^2 \quad (24)$$

$$V_{\text{frame}} = \pi \cdot \left((d_{\text{pack}})^2 - (d_{\text{face}})^2 \right) \cdot \frac{t_{\text{frame}}}{4} = 3.142 \cdot ((70.000 \text{ mm})^2 - (58.000 \text{ mm})^2) \cdot \frac{0.200 \text{ mm}}{4} = 241.274 \text{ mm}^3 \quad (25)$$

$$m_{\text{housings}} = 2 \cdot V_{\text{half}} \cdot \rho_{\text{ti}} = 2 \cdot 33552.778 \text{ mm}^3 \cdot 4510.000 \frac{\text{kg}}{\text{m}^3} = 302.646 \text{ g} \quad (26)$$

$$m_{\text{frames}} = 2 \cdot V_{\text{frame}} \cdot \rho_{\text{ss}} = 2 \cdot 241.274 \text{ mm}^3 \cdot 8000.000 \frac{\text{kg}}{\text{m}^3} = 3.860 \text{ g} \quad (27)$$

$$m_{\text{coarse}} = 2 \cdot A_{\text{pack}} \cdot w_{\text{coarse}} = 2 \cdot 3848.451 \text{ mm}^2 \cdot 1.367 \frac{\text{kg}}{\text{m}^2} = 10.522 \text{ g} \quad (28)$$

$$m_{\text{fine}} = A_{\text{pack}} \cdot w_{\text{fine}} = 3848.451 \text{ mm}^2 \cdot 0.950 \frac{\text{kg}}{\text{m}^2} = 3.656 \text{ g} \quad (29)$$

$$m_{\text{fine}} = 3.656 \text{ g}$$

DRY FINISHED MASS WITH A DEVELOPMENT RESERVE

$$m_{\text{dry}} = m_{\text{housings}} + m_{\text{frames}} + m_{\text{coarse}} + m_{\text{fine}} = 302.646 \text{ g} + 3.860 \text{ g} + 10.522 \text{ g} + 3.656 \text{ g} = 0.707 \text{ lb} \quad (30)$$

$$m_{\text{reserve}} = \text{growth} \cdot m_{\text{dry}} = 1.200 \cdot 0.707 \text{ lb} = 0.848 \text{ lb} \quad (31)$$

$$\text{MS}_{\text{mass}} = \frac{m_{\text{limit}}}{m_{\text{reserve}}} - 1 = \frac{1.000 \text{ lb}}{0.848 \text{ lb}} - 1 = 0.179 \quad (32)$$

$$\text{MS}_{\text{mass}} = 0.179$$

MASS SCOPE

[insert text here]

24 / REFERENCES

[insert text here]

REFERENCES

- [1] RPD HIFLO-S / 5-S catalog row, Haver & Boecker — Accessed 2026-09-14. 5-S: 850 x 380, 5 μ m geometric pore, zeta 9020, 0.18 mm, 0.95 kg/m², 34% porosity. No beta-3 certification or applicable water flow curve supplied. oh.haverboecker.com/woven-wire-filters/weave-ty...
- [2] Woven expertise / Dutch twilled weave, GKD, Printed p.14 — 5–250 μ m absolute opening family; 304L/316L materials. Exact 5 μ m construction and beta-3 performance require RFQ. www.gkd-group.com/wp-content/uploads/2024/09/br...
- [3] 040X040T0100 / 40 mesh T316, 0.010 inch wire, TWP Inc. — 0.254 mm wire, 0.381 mm opening, 36% open area, 0.508 mm thickness, 0.280 lb/ft². Catalog alloy is 316, not certified 316L. www.twpinc.com/40-mesh-t316-stainless-010-wire...
- [4] 40316.010PL / 40 x 40 plain weave, Edward J. Darby & Son — Second exact-construction coarse supplier: 316, 0.010 inch wire, 0.015 inch opening, 36% open area. 316L requires separate certification. catalog.darbywiremesh.com/item/filtration-and-s...
- [5] Wire cloth and filter fabrication capabilities, Dorstener Wire Tech — Resistance welding, laser cutting and cleaning capabilities. Exact fine cloth, closed seam and lot origin not yet quoted. dwt-inc.com
- [6] TIMETAL 50A / CP titanium Grade 2, TIMET — Density 4.51 g/cm³; guaranteed 275 MPa minimum yield; typical modulus range 105–120 GPa. Do not use typical tensile strength as a minimum. www.timet.com/documents/datasheets/cp-grades/ti...
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- [8] Supra range / 316L ASTM A240 properties, Outokumpu, Table 6 — 170 MPa minimum yield and 485 MPa minimum tensile for listed 316L sheet/plate forms; not a woven-wire allowable. www.outokumpu.com/-/media/files/products/supra...
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- [10] Orbital Welding User's Manual, Swagelok — Process and shielding terminology; no prequalified F100 weld schedule is claimed. www.swagelok.com/downloads/WebCatalogs/en/MS-13...
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- [12] Supplementary Release on Properties of Liquid Water, IAPWS, 2011 — Ambient-pressure property approximation at 70 F; final qualification should account for pressure dependence. www.iapws.org/relguide/LiquidWater.pdf
- [13] ASTM F316 pore characterization, ASTM International — Membrane scope; a woven-metal method needs applicability review. Pore characterization alone does not establish particle retention. store.astm.org/standards/f316
- [14] ISO 16889:2022 / multi-pass filter evaluation, ISO — Method reference, not a claim that the proposed water/3 μ m challenge complies with this hydraulic-fluid standard. www.iso.org/standard/77245.html
- [15] Garrett / project input, User-supplied design brief and pressure-drop equations, 2026 — Governing architecture and equations supplied in task. Clarification: housing gauge pressure, not blocked-element differential; water at 70 F. Original NASA report identifier/file not supplied, so its table entries are treated as user-transcribed data.